Deduce the World: How Far Can Reasoning Go?

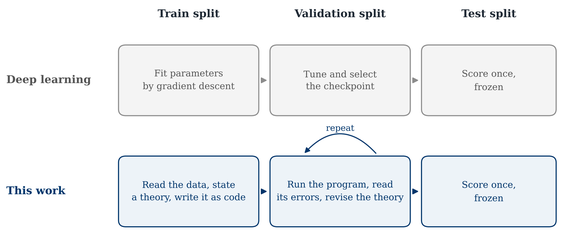
Jeffrey Wei — Yale University
Brian Cheung — University of California, San Francisco

Results & figures

1 Theories, not fitted predictors

Science deduces the world from first principles, revising its theories where their predictions fail.

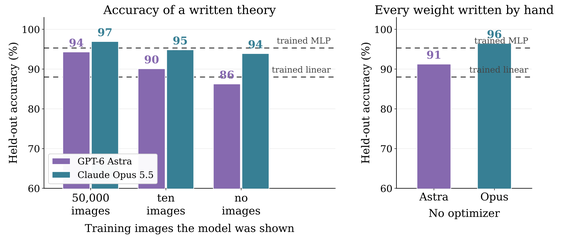
We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.



May	read papers and the web, measure the training data, inspect its own errors
May not	fit parameters to labels, search automatically over rules, store examples, see the test split

2 How much data a theory needs

MNIST: 60,000 training and 10,000 test images of handwritten digits.

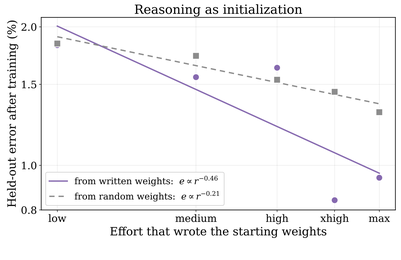


Best of five reasoning efforts. Dashed lines are trained models.

> Given no training images at all, a theory of how digits are written classifies most of them.

Reasoning as initialization

A written theory is also a starting point. Networks initialized from written weights train to lower error, and the gap widens with reasoning effort.



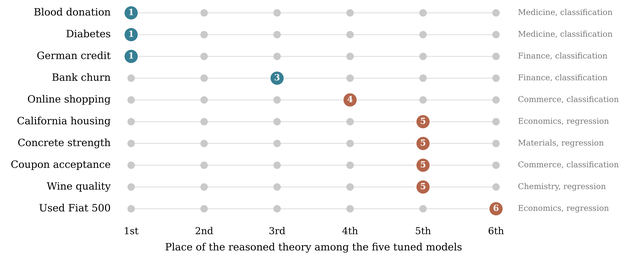
3 The same protocol, ten fields

Ten tasks from TabArena, a benchmark of real tabular problems with fixed splits and tuned baselines.

What a theory looks like

From one program on German credit. The points were stated before measurement, not fitted.

Checking account	none $+3$, at least 200 DM $+1$, negative -2
Credit history	critical account elsewhere $+2$, past delay -1
Loan duration	-1 for each twelve months beyond the first twelve
Output	$P(\text{good}) = 1/(1+e^{-(\text{score}-1)/2})$

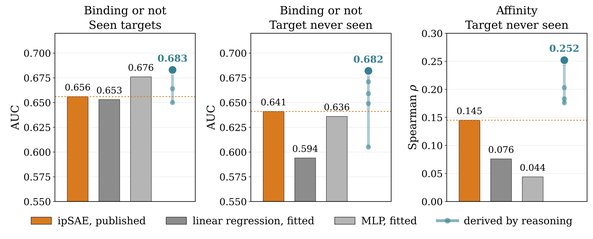


Each task ranks the theory against five tuned models on the same rows.

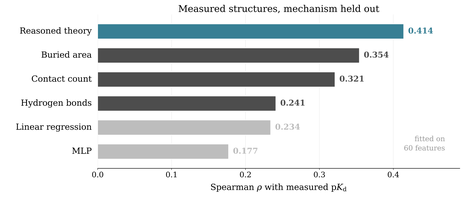
> Theories win on small clinical and credit tables, and trail boosted trees where the signal lies in feature interactions.

4 Deriving a scientific score

Each session receives a structure predictor's error matrix and nothing else.



704 designed binders. The right two panels hold out a whole target.



One mechanism of association held out whole.

> Every unfitted physical quantity beats both fitted models, and the reasoned rule beats all of them.

What a theory looks like

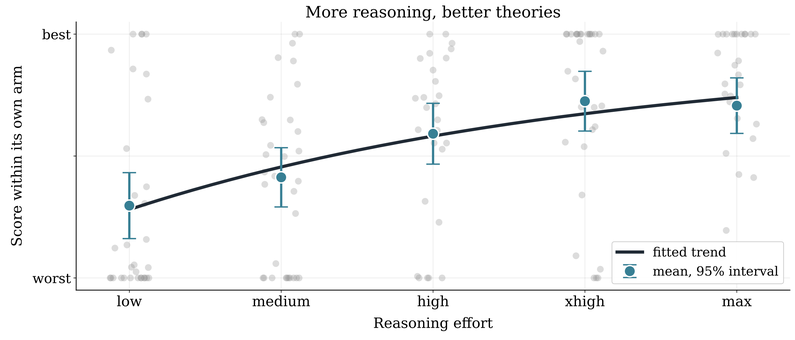
The chain one session stated before writing any code.

Harmonic well	Equipartition sets the force constant to the thermal energy over the square of the fluctuation.
Stiffness is depth	Curvature at the minimum is proportional to depth. A stiffer interface is a deeper one.
Divide out the predictor	Measuring the interface error against the binder's own internal error removes predictor quality, and the units with it.
Count what binding confines	Association removes six rigid-body degrees of freedom, each contributing half the thermal energy times the log of the stiffness.

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The three is six halved. Nothing is fitted.

5 Scaling reasoning effort



Every arm on one axis, each normalized to its own range.

> More reasoning makes better theories, with no gain past the fourth setting.

6 Hypothesis

Deep learning dominates the domains that are data-rich and theory-poor. We hypothesize that as reasoning improves, language models will supply the first principles those domains lack.